

IP as the IoT Network Layer

Dr. Abbas Javed

Assistant Professor

Department of Electrical and Computer Engineering,

COMSATS University Islamabad, Lahore Campus

abbasjaved@cuilahore.edu.pk

Contents

- **The Business Case for IP:** This section discusses the advantages of IP from an IoT perspective and introduces the concepts of adoption and adaptation.
- **The Need for Optimization:** This section dives into the challenges of constrained nodes and devices when deploying IP. This section also looks at the migration from IPv4 to IPv6 and how it affects IoT networks.
- **Optimizing IP for IoT:** This section explores the common protocols and technologies in IoT networks utilizing IP, including 6LoWPAN, 6TiSCH, and RPL.

Key Advantages of Internet Protocol

- Open and standards-based
- Versatile
- Ubiquitous
- Scalable
- Manageable and highly secure
- Stable and resilient
- Consumers' market adoption
- The innovation factor

Adoption or Adaptation of the Internet Protocol

- ***Adaptation*** means application layered gateways (ALGs) must be implemented to ensure the translation between non-IP and IP layers.
- ***Adoption*** involves replacing all non-IP layers with their IP layer counterparts, simplifying the deployment model and operations.

Factors for determining the model

- Bidirectional versus unidirectional data flow
- Overhead for last-mile communications paths (require optimization)
- Data flow model
- Network diversity

The Need for Optimization

- Constrained Nodes
 - “thing” architecture may or may not offer similar characteristics compared to generic PC or server in an IT environment
 - Unreliable communication path
 - Power consumption
- **Devices that are very constrained in resources, may communicate infrequently to transmit a few bytes, and may have limited security and management capabilities:** This drives the need for the IP adaptation model, where nodes communicate through gateways and proxies.
- **Devices with enough power and capacities to implement a stripped-down IP stack or non-IP stack:** In this case, you may implement either an optimized IP stack and directly communicate with application servers (adoption model) or go for an IP or non-IP stack and communicate through gateways and proxies (adaptation model).
- **Devices that are similar to generic PCs in terms of computing and power resources but have constrained networking capacities, such as bandwidth:** These nodes usually implement a full IP stack (adoption model), but network design and application behaviors must cope with the bandwidth constraints.

Optimizing IP for IoT

- Constrained nodes and constrained networks mandate optimization at various layers and on multiple protocols of the IP architecture.

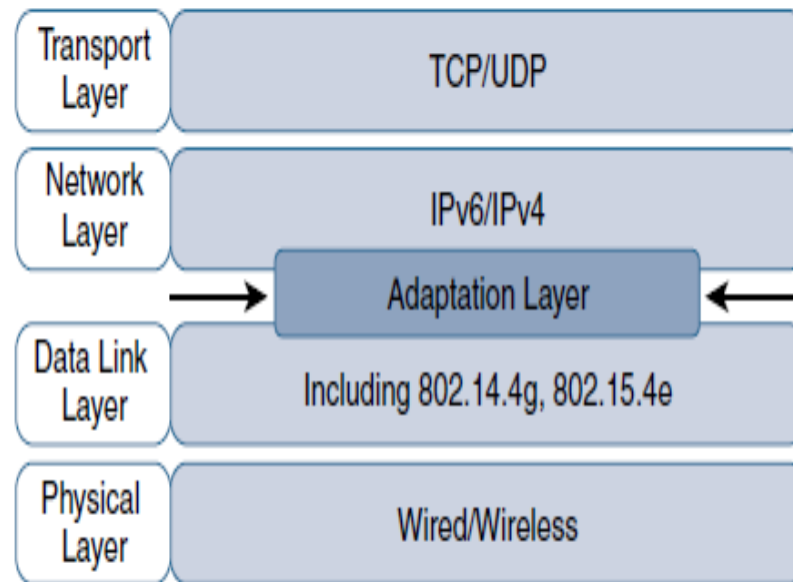


Figure 5-1 *Optimizing IP for IoT Using an Adaptation Layer*

From 6LoWPAN to 6Lo

- The initial focus of the 6LoWPAN working group was to optimize the transmission of IPv6 packets over constrained networks such as IEEE 802.15.4

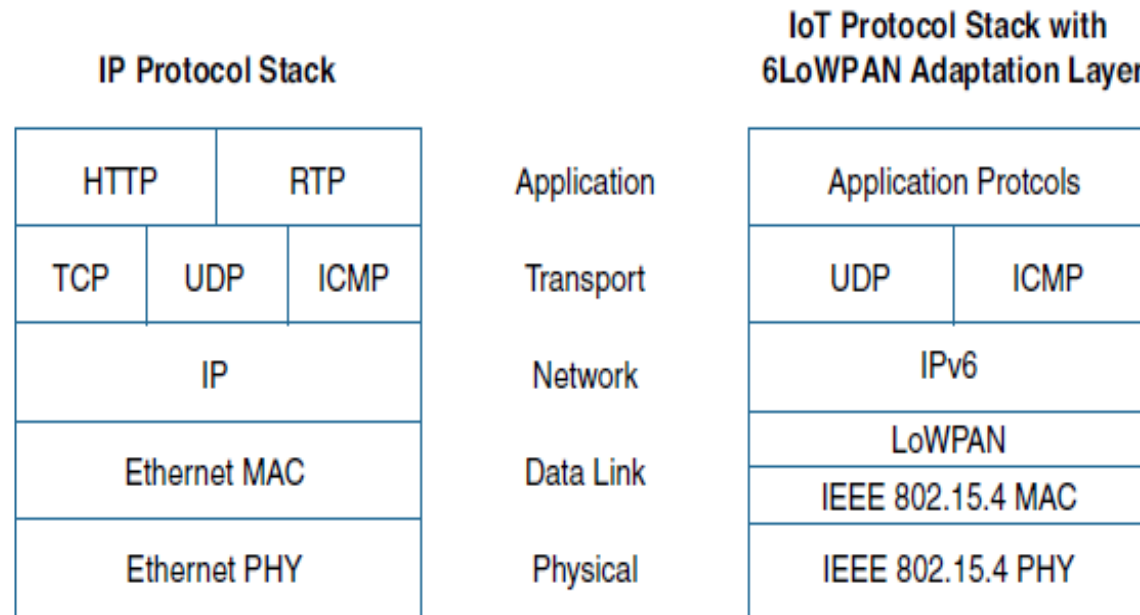


Figure 5-2 Comparison of an IoT Protocol Stack Utilizing 6LoWPAN and an IP Protocol Stack

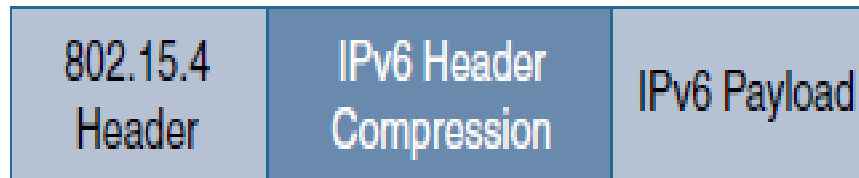


Figure 5-3 *6LoWPAN Header Stacks*

- The 6LoWPAN working group published several Request for comments (RFCs), but RFC 4994 is foundational because it defines frame headers for the capabilities of header compression, fragmentation, and mesh addressing. These headers can be stacked in the adaptation layer to keep these concepts separate while enforcing a structured method for expressing each capability.

Header Compression

- IPv6 header compression for 6LoWPAN was defined initially in RFC 4944 and subsequently updated by RFC 6282. This capability shrinks the size of IPv6's 40-byte headers and User Datagram Protocol's (UDP's) 8-byte headers down as low as 6 bytes combined in some cases.
- header compression for 6LoWPAN is only defined for an IPv6 header and not IPv4.
- At a high level, 6LoWPAN works by taking advantage of shared information known by all nodes from their participation in the local network
- it omits some standard header fields by assuming commonly used values

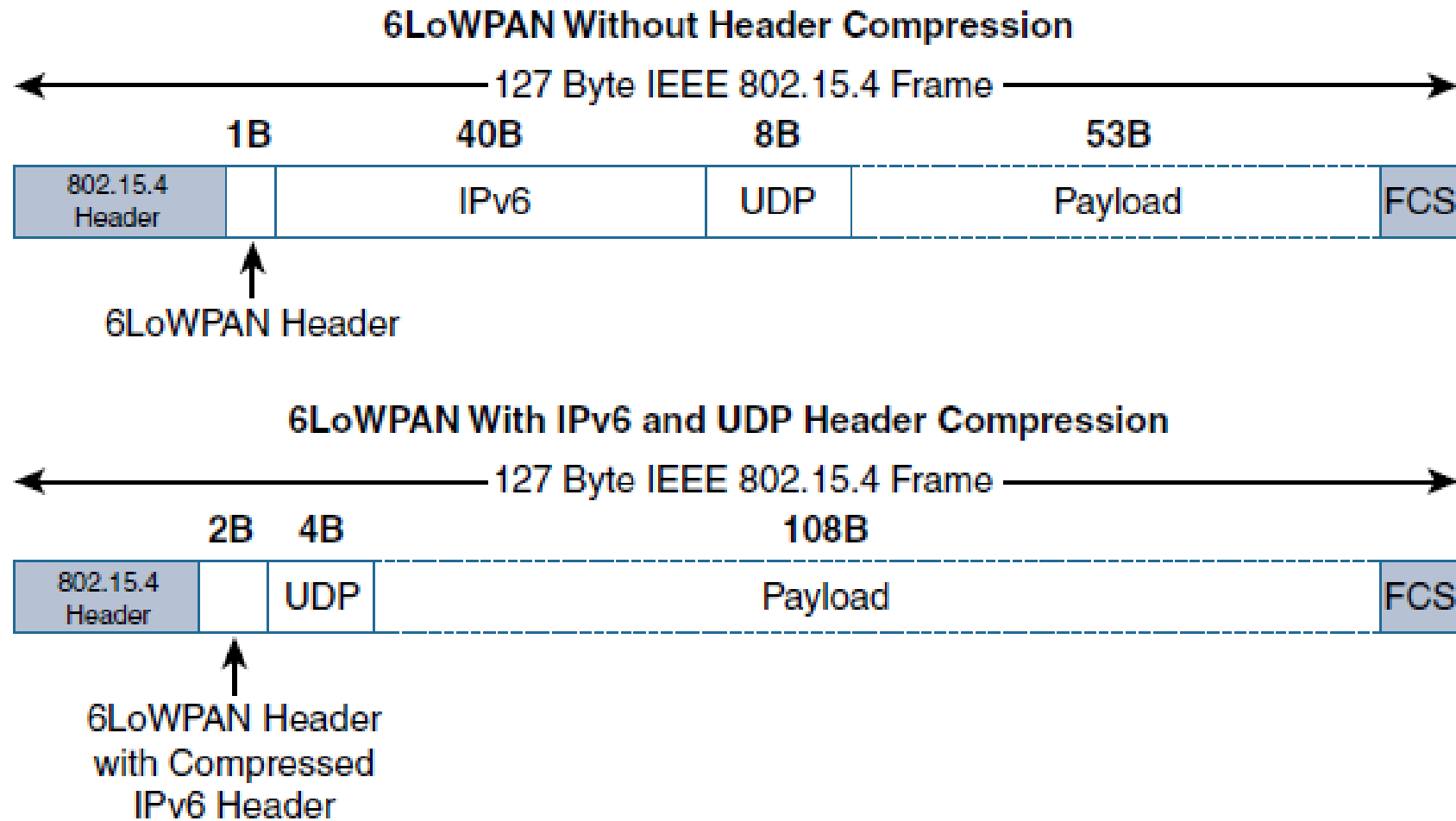


Figure 5-4 *6LoWPAN Header Compression*

Fragmentation

- The maximum transmission unit (MTU) for an IPv6 network must be at least 1280 bytes.
- For IEEE 802.15.4, 127 bytes is the MTU.
- You can see that this is a problem because IPv6, with a much larger MTU, is carried inside the 802.15.4 frame with a much smaller one.
- To remedy this situation, large IPv6 packets must be fragmented across multiple 802.15.4 frames at Layer 2.

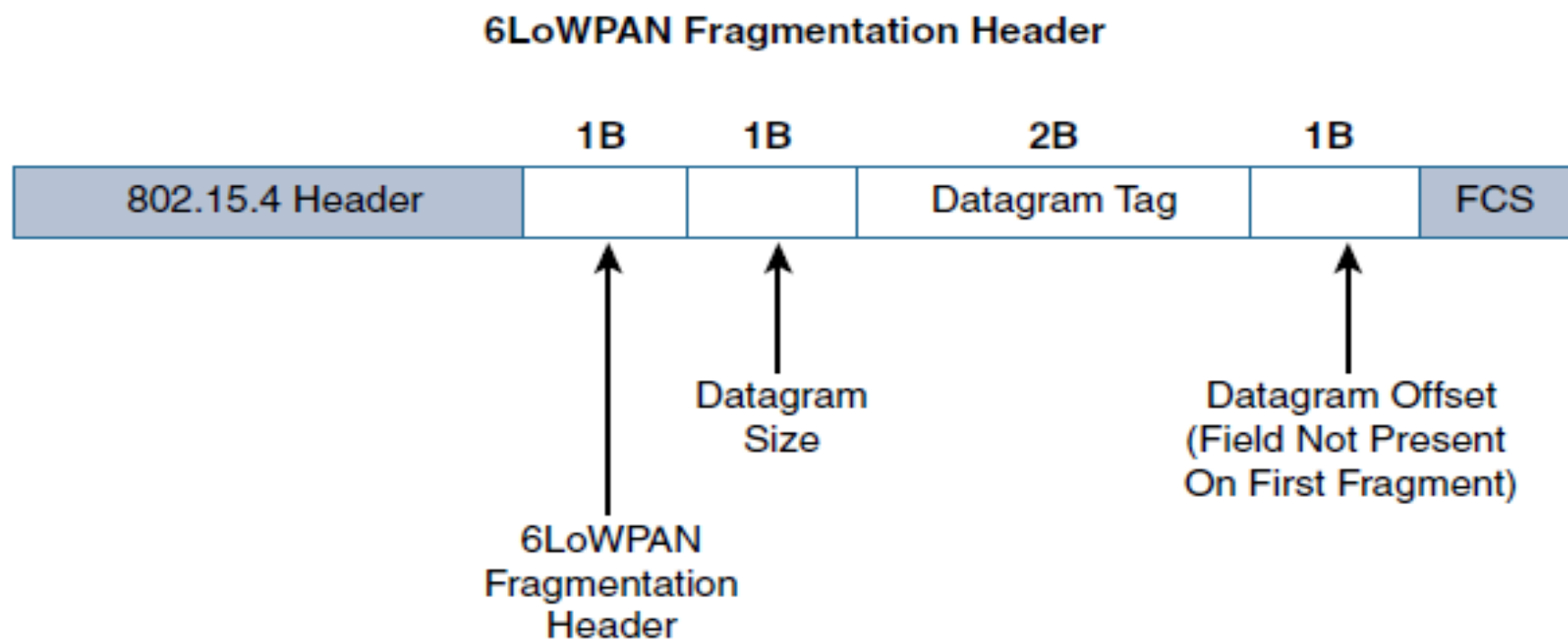


Figure 5-5 *6LoWPAN Fragmentation Header*

From the Libr

- The fragment header utilized by 6LoWPAN is composed of three primary fields: Datagram Size, Datagram Tag, and Datagram Offset
- The 1-byte Datagram Size field specifies the total size of the unfragmented payload.
- Datagram Tag identifies the set of fragments for a payload.
- Finally, the Datagram Offset field delineates how far into a payload a particular fragment occurs.

Mesh Addressing

- The purpose of the 6LoWPAN mesh addressing function is to forward packets over multiple hops
- Three fields are defined for this header: Hop Limit, Source Address, and Destination Address
- Analogous to the IPv6 hop limit field, the hop limit for mesh addressing also provides an upper limit on how many times the frame can be forwarded
- Each hop decrements this value by 1 as it is forwarded. Once the value hits 0, it is dropped and no longer forwarded.
- The Source Address and Destination Address fields for mesh addressing are IEEE 802.15.4 addresses indicating the endpoints of an IP hop

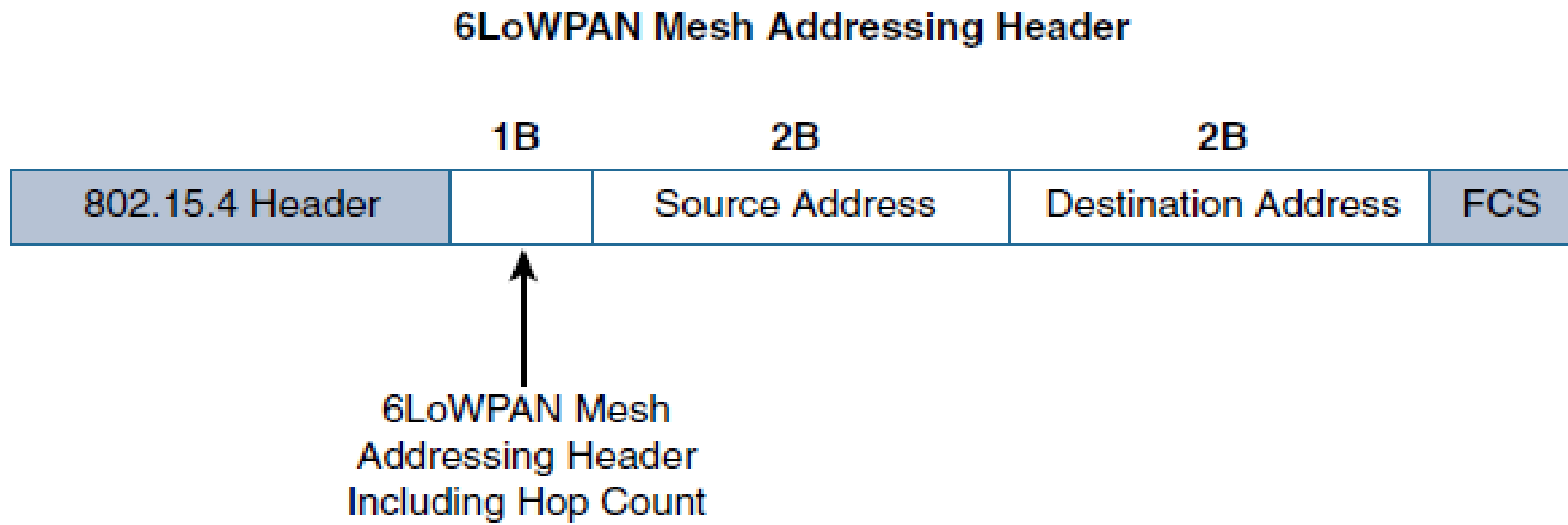


Figure 5-6 *6LoWPAN Mesh Addressing Header*

6LowPAN Packet

IEEE 802.15.4 DATA #74

From 0xABCD/00:12:74:02:00:02:02:02

to 0xABCD/00:12:74:06:00:06:06:06

Sec = false, Pend = false, ACK = true, iPAN = true,

DestAddr = Long, Vers. = 1, SrcAddr = Long

IPHC HC-06 TF = 3, NH = compressed, HLIM = 64, CID = 1,

SAC = stateful, SAM = 3, MCast = false, DAC = stateful, DAM = 3

Contexts: sci=0 dci=0

IPv6 TC = 0, FL = 0, ttl = 64

From aaaa:0000:0000:0000:0012:7402:0002:0202

to aaaa:0000:0000:0000:0012:7406:0006:0606

Frag Header nth size = 262, tag = 0x6304, offset = 0

Payload (17 bytes) (hex dump)

00 01 16 F0 22 3D 16 2E 5E 34 68 65 6C 6C 6F 20 37

ASCII (partial): ...^4hello 7

MAC layer (IEEE 802.15.4)

- Frame source & destination:
- PAN ID = 0xABCD (both endpoints use same PAN — a Personal Area Network).
- Source MAC = 00:12:74:02:00:02:02:02 (64-bit long address).
- Destination MAC = 00:12:74:06:00:06:06:06 (64-bit).
- Flags:Sec = false → no 802.15.4 MAC security (no MAC-layer encryption/auth).
- ACK = true → acknowledgements are enabled; sender expects an ACK from receiver.
- iPAN = true → indicates an intra-PAN packet (PAN ID compression may be used).
- DestAddr = Long and SrcAddr = Long → 64-bit addresses used (not short 16-bit).

6LoWPAN IPHC (IPv6 Header Compression)

- TF = 3
 - Traffic Class & Flow Label compression indicator. TF=3 here means Traffic Class and Flow Label are elided (they are not carried in the packet) — the receiver must assume default values (TF = 0/0) when reconstructing IPv6 header.
- NH = compressed
 - Next Header is compressed (i.e., a NHC — Next Header Compression — is used). This often means the next header is UDP (and UDP is compressed using LOWPAN_NHC_UDP) or another compressed protocol.
- HLIM = 64
 - The Hop Limit (TTL) value for IPv6 packets is 64 (explicitly present or indicated). CID = 1A Context Identifier Extension is present — a 6LoWPAN context is in use. Contexts allow IPv6 prefixes to be replaced by a short ID to compress addresses.
- SAC / DAC = stateful;
 - SAM / DAM = 3 These say that source/destination addressing uses stateful (context-based) compression and the addressing mode value (SAM/DAM) = 3. In practice this means the full interface identifier (IID) is derived from the link-layer address or from a context, not carried as 128-bit inline address. (In your packet the decompressed IPv6 addresses are shown, so the receiver used MAC/context info to reconstruct them.)

Reconstructed IPv6 addresses

- The packet lists the reconstructed IPv6 addresses:
 - Source IPv6: aaaa:0000:0000:0000:0012:7402:0002:0202
 - Destination IPv6: aaaa:0000:0000:0000:0012:7406:0006:0606
- The low 64 bits of the IPv6 addresses (the Interface Identifier / IID) match the source/dest 64-bit MACs shown at MAC-layer (the 0012:7402:0002:0202 part looks like the 64-bit MAC mapped directly as IID). Thus the compressor elided the IPv6 IID and the receiver reconstructed it from the MAC or context.

Fragmentation header

- Frag Header nth size = 262, tag = 0x6304, offset = 0
 - This is a 6LoWPAN fragmentation header.
 - size = 262 → the full IPv6 datagram length (262 bytes) before fragmentation.
 - tag = 0x6304 → a tag to group fragments belonging to the same IPv6 datagram.
 - offset = 0 → this is the first fragment (offset zero).
 - The actual payload here is only 17 bytes because this is the first fragment; other fragments (if present) will contain the rest of the 262-byte IPv6 datagram.